

Problem Statement: Building a Scalable Test Automation Strategy for a Multi-Tenant SaaS Healthcare Platform

Background:

You are working with a SaaS healthcare platform used by multiple hospitals and clinics. The platform provides core patient management features, and each tenant (hospital/clinic) has its own set of custom modules and integrations (e.g., lab systems, insurance providers).

The platform is built on a microservices architecture, but many custom modules are still tightly coupled and lack clear API boundaries.

There are 10 scrum teams: 6 focused on tenant-specific customizations, 3 on platform services, and 1 on integrations.

Each team has 2 developers and 1 QA engineer.

Releases are deployed monthly, but hotfixes are frequent due to production issues.

50% of production incidents are related to integration failures or data mismatches between modules.

Test automation exists for platform services (API level), but not for custom modules or integrations.

Manual regression testing is performed by QA engineers, but coverage is inconsistent and not tracked.

No unified test data management strategy exists; each team creates its own test data ad hoc.

Quality metrics are collected only for platform services, not for custom modules or integrations.

Task:

Propose a comprehensive test automation and data management strategy that ensures end-to-end quality across both platform and tenant customizations, with a focus on integration reliability and scalable regression coverage.